

View Reviews

Paper ID

341

Paper Title

Heterogeneous Multicore SoC Architecture for Real-Time Biomedical Data Analysis

Reviewer #1

Questions

1. Relevance and timeliness: rate the paper content's relevance to the conference theme.

Good

3. Novelty and originality: Please comment on the novelty and significant contributions addressed in the paper within the area of research.

Strength: Its a survey paper. The paper addresses an important and emerging research area—hardware security and silicon-level trust in edge AI systems. The topic is timely due to the rapid deployment of AI accelerators, chiplets, and heterogeneous integration in modern computing platforms. The authors attempt to survey vulnerabilities across the semiconductor lifecycle and discuss defensive architectures.

4. Rate the comprehensiveness and relevance of the literature review.

Good

5. Technical content and scientific presentation: rate the presented technical results (methodology, simulation, practical analysis, and limitations).

Results are not sufficient in the area of intended research.

6. Quality of presentation: rate the paper's structure and clarity of presentation.

Average

7. What are the positive and weak aspects of the paper?

Positive-The paper focuses on a relevant research topic involving hardware vulnerabilities in edge AI systems.

It attempts to analyze threats across multiple stages of the semiconductor lifecycle, including design, manufacturing, and deployment.

The taxonomy of vulnerabilities provides a structured way to categorize threats such as hardware Trojans, side-channel attacks, and supply-chain risks.

The survey includes relatively recent literature and attempts to highlight research gaps.

8. Recommended changes in the manuscript (if any)

The paper lacks strong analytical depth. Many sections simply summarize existing works without critically comparing them or explaining their limitations.

The novelty of the survey is not clearly established. Several existing surveys already discuss hardware security, hardware Trojans, and supply-chain vulnerabilities. The paper does not clearly explain how it differs from or improves upon those surveys.

The evaluation framework proposed by the authors is not fully applied to the surveyed literature. A stronger paper would include a comprehensive comparison of prior work using this framework.

The discussion of defensive architectures is relatively brief and lacks detailed analysis of design trade-offs, scalability, and implementation challenges.

The literature coverage could be expanded. Some well-known work on chiplet security, AI accelerator leakage, and hardware Trojan detection appears to be missing or only briefly mentioned.

10. Do you recommend this manuscript for extended journal publication?

Yes

Reviewer #5

Questions

1. Relevance and timeliness: rate the paper content's relevance to the conference theme.

Good

2. Novelty and Originality: Rate the level of novelty and originality of the proposed work compared to existing literature.

Strong novelty with clear improvement over existing work

3. Novelty and originality: Please comment on the novelty and significant contributions addressed in the paper within the area of research.

Current healthcare monitoring systems often send biomedical data to the cloud or rely on general-purpose processors, which can introduce delays in analysis and response. For critical applications (e.g., arrhythmia detection, seizure prediction), these delays may be unacceptable, so the work motivates moving all computation on-chip to reduce latency and enable true real-time monitoring.

4. Rate the comprehensiveness and relevance of the literature review.

Good

5. Technical content and scientific presentation: rate the presented technical results (methodology, simulation, practical analysis, and limitations).

Results are good and presented well.

6. Quality of presentation: rate the paper's structure and clarity of presentation.

Very good

7. What are the positive and weak aspects of the paper?

Demonstrate that integrating domain-specific processing (DSP core, custom instructions, and a hardware CNN accelerator) on-chip can simultaneously improve latency, power efficiency, and data privacy compared with cloud-dependent or general-purpose-processor-based healthcare monitoring systems.

8. Recommended changes in the manuscript (if any)

Include the readable Figure 2. For a proposed architecture, give the figure number. No comparison is shown between the proposed work with existing work.

9. Overall Recommendation:

Accept with minor revisions

10. Do you recommend this manuscript for extended journal publication?

No